Stage 16N Inhomogeneous Plate-with-Hole Cycle-Jump Benchmark
Living Technical Report

Master Thesis Preparation

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Abstract

This living report records the geometry, numerical setup, HPC configuration, reference results, reinjection tests, diagnostic failures, and current conclusions for Stage 16N. The stage uses an inhomogeneous Abaqus plate-with-hole benchmark with a NEML-equivalent Chaboche UMAT. The full 1000-cycle non-jump reference has been completed and is used as the validation baseline for later fixed and adaptive cycle-jump simulations. The current fixed-jump conclusion is frozen at a narrow accepted cycle-100 jump, `B1D5`/`B1D5_EQ`: 100 $\rightarrow$ 105 $\rightarrow$ 250 with 2.34986% maximum primary scalar-metric error. Larger and later manually initialized jumps remain blocked because `SDVINI`/`SIGINI`-based reinjection does not robustly reconstruct the later-cycle inhomogeneous finite-element state.

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1 Purpose and Current Status

The purpose of Stage 16N is to move from single-element or homogeneous cycle-jump demonstrations to an inhomogeneous finite-element benchmark. The model is deliberately more demanding: a plate with a central hole produces localized cyclic plasticity near the hole, so the method must be judged not only from global reaction-force loops but also from local stress and state-variable evolution.

Table 1: Current Stage 16N infrastructure and validation status.

Item	Status
1000-cycle full non-jump reference	Completed
16-CPU production policy	Locked
Abaqus threaded launcher	Verified
Adaptive ΔN table	Completed
Exact reinjection scaffold	Prepared
HPC mail policy	Fixed
B0-1 exact reinjection solver run	Completed
B0-1 scientific result	Practical baseline accepted
B0 initialization-only audit	Completed
B0-2 / B0-3 exact reinjection jobs	On hold
Fixed state-initialized jump validation	Scaffold prepared
Adaptive cycle-jump workflow	On hold

2 Geometry and Model Definition

The Stage 16N model is a three-dimensional plate-with-hole benchmark. It is intended to create localized cyclic evolution near the hole while retaining a global hysteresis response that can be compared across full-reference, exact-reinjection, fixed-jump, and adaptive-jump runs.

Table 2: Stage 16N model geometry and mesh summary.

Quantity	Value
Plate length	20.0
Plate height	10.0
Plate thickness	1.0
Central hole radius	1.0
Mesh spacing	$\Delta x = \Delta y = 0.25$
Element type	C3D8
Nodes	6642
Elements	3148
Hole-ring elements	60
Material model	NEML-equivalent Chaboche UMAT
State variables	27 SDVs

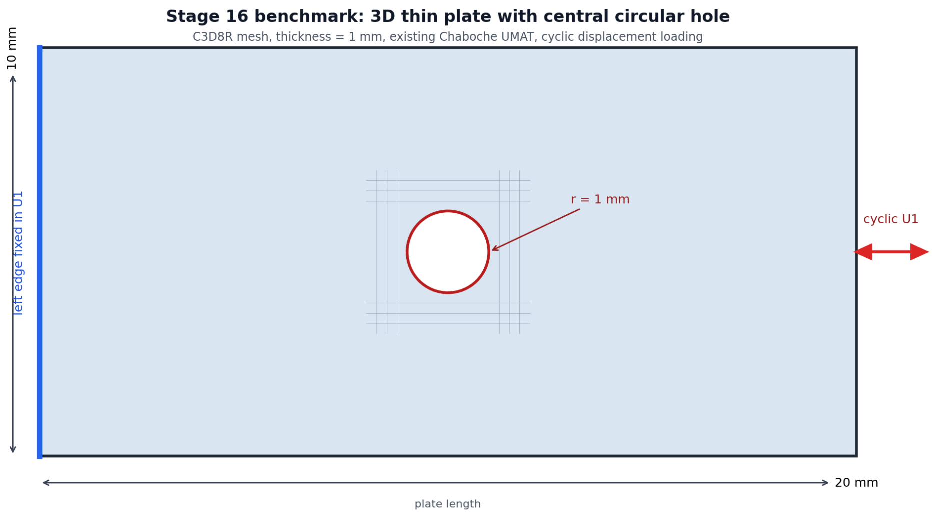

Figure 1: Plate-with-hole geometry used for the Stage 16N benchmark.

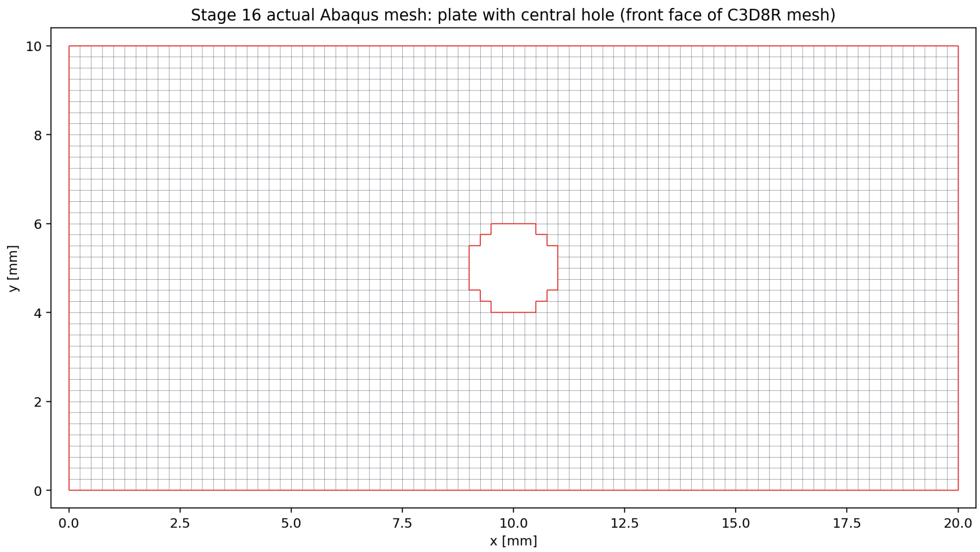

Figure 2: Actual structured C3D8 mesh with the element-wise approximation of the central hole.

2.1 Boundary Conditions and Loading

The left edge is fixed in the axial direction. Two anchor nodes constrain rigid-body modes. The right edge receives a cyclic axial displacement amplitude of ± 0.10. The full reference contains 1000 full non-jump cycles. Selected field output is stored at cycles 1, 2, 10, 50, 100, 250, 500, 750, and 1000.

3 HPC and Production Policy

The first corrected 1000-cycle parallel reference used 30 OpenMP threads, but PBS accounting showed that the average effective CPU usage was far below the requested 30 cores. A 16-CPU verification run was therefore used to lock a more resource-efficient production policy.

Table 3: Locked Stage 16N production setting.

Setting	Value
Production default	1 MPI rank × 16 OpenMP threads
PBS request	`select=1:ncpus=16:mpiprocs=1:ompthreads=16`
Abaqus launch	`cpus=16 mp_mode=threads`
Expected message-file line	`1 MPI RANK x 16 THREADS`
Maximum walltime per job	24:00:00
Maximum simultaneous jobs	2
Maximum active Stage 16N cores	32

This policy is not claimed to perfectly saturate all 16 CPUs. It is a resource-efficient compromise that prevents the earlier serial fallback and avoids reserving 30 CPUs for a workload that did not use them efficiently.

3.1 Two-Job Scheduling Policy

From 2026-06-06 onward, Stage 16N solver planning assumes at most two simultaneous production jobs. Each job requests 16 CPU cores with one MPI rank and 16 OpenMP threads, so the intended active usage is 32 CPU cores. The two slots are used only after the relevant method gate has passed. First-time method checks, such as the first fixed cycle-jump case B1, run alone until extracted and compared. If B1 passes, B2 and B3 can be submitted together. If B1 fails badly, B2 and B3 remain held and the two slots are used instead for smaller or diagnostic B1 variants.

3.2 Mail Notification Policy

Future PBS jobs must explicitly request mail on abort, begin, and end:

```
#PBS -m abe
#PBS -M pr21vyci@mailserver.tu-freiberg.de
```

This was added because PBS accounting previously showed `Mail_Points = a`, which only requests abort/failure mail.

4 Full 1000-Cycle Non-Jump Reference

The full non-jump reference was completed using Abaqus/Standard with the corrected threaded launcher.

Table 4: Completed 1000-cycle reference run.

Quantity	Value
PBS job	1335408.mmaster02
Abaqus job	`stage16n_parallel_max_reference_1000cycles`
Walltime	17:56:43
CPU time	178:03:42
Requested resources	30 CPUs, 135 GB
Abaqus parallelism	1 MPI rank × 30 threads
Completed cycles	1000
Last partial cycle	none

Table 5: Reference evolution from cycle 1 to cycle 1000.

Quantity	Cycle 1	Cycle 1000	Change
RF1_max	2356.834	2908.244	+23.40%
RF1_min magnitude	2509.712	3074.823	+22.52%
loop_area_abs	433.822	577.068	+33.02%
HOLE_RING_MISES_MAX	398.503	492.827	+23.67%
HOLE_RING_SDV1_MAX	0.0561	26.0519	–
HOLE_RING_SDV8_MAX	0.2996	92.4455	–
HOLE_RING_SDV11_MAX	7.9019	61.3076	–

The evolution clearly exceeds the original continuation thresholds for global loop quantities and local hole-ring state variables. This confirms that the plate-with-hole benchmark is meaningful for cycle-jump validation.

5 Adaptive ΔN Table

The adaptive ΔN table was computed from the completed 1000-cycle reference. It uses dense global cycle metrics and selected local field-output anchors. The controlling variable is the most sensitive monitored quantity, not only the global reaction force.

Table 6: Conservative adaptive ΔN estimates from the 1000-cycle reference.

Base cycle	Next anchor	Target	ΔN	Controlling variable
1	2	2	1	HOLE_RING_SDV8_MAX
2	10	3	1	HOLE_RING_SDV8_MAX
10	50	14	4	HOLE_RING_SDV8_MAX
50	100	61	11	HOLE_RING_SDV1_MAX
100	250	124	24	HOLE_RING_SDV1_MAX
250	500	299	49	HOLE_RING_SDV1_MAX
500	750	575	75	HOLE_RING_SDV1_MAX
750	1000	849	99	HOLE_RING_MISES_MAX

The table shows that local state variables near the hole strongly restrict the allowable jump size. This supports the decision that any later fixed/adaptive jump validation must include local metrics, not only global RF loops.

5.1 Reinjection-Aware Controller Update

After the B0 initialization audit, HOLE_RING_SDV8_MAX was downgraded from a hard pass/fail controller to a diagnostic-only variable because it already shows a large mismatch immediately after manual SDVINI/SIGINI initialization. A reinjection-aware table was therefore generated with SDV8 retained in diagnostic output but removed from the hard controller list.

Table 7: Reinjection-aware adaptive ΔN estimates with SDV8 diagnostic-only.

Base	Next anchor	Target	ΔN	Controller
1	2	2	1	HOLE_RING_SDV11_MAX
2	10	3	1	HOLE_RING_SDV1_MAX
10	50	15	5	HOLE_RING_SDV1_MAX
50	100	61	11	HOLE_RING_SDV1_MAX
100	250	124	24	HOLE_RING_SDV1_MAX
250	500	299	49	HOLE_RING_SDV1_MAX
500	750	575	75	HOLE_RING_SDV1_MAX
750	1000	849	99	HOLE_RING_MISES_MAX

The first conservative fixed-jump gate remains cycle 100 to cycle 125, followed by normal continuation to cycle 250. This row is still controlled by HOLE_RING_SDV1_MAX, not by diagnostic-only SDV8.

6 Stage 16N-B: State-Initialized Continuation Verification

Before any real cycle jump is allowed, the workflow must prove that stress and state-variable fields can be transferred into a new Abaqus run. The route used here is manual state-initialized continuation through SDVINI/SIGINI; it is not a native Abaqus restart. The originally intended verification cases were:

- B0-1: exact cycle 100 state, continue normally to cycle 250.
- B0-2: exact cycle 250 state, continue normally to cycle 500.
- B0-3: exact cycle 500 state, continue normally to cycle 1000.

B0-2 and B0-3 remain on hold. The B0-1 and initialization-audit results are now treated as the measured reinjection baseline rather than as proof of an exact local restart.

6.1 State Reader Implementation

The initial CSV preload design failed under threaded Abaqus because multiple threads attempted to load the same file concurrently. The state-transfer implementation was therefore changed to a direct-access binary reader. Each record contains:

```
S1, S2, S3, S4, S5, S6, SDV1, ..., SDV27
```

The record number is:

```
record = (NOEL - 1) * 8 + NPT
```

On the Abaqus/Intel Fortran environment used here, direct-access RECL is interpreted in 4-byte words. Therefore the correct value for 33 double-precision values is:

```
RECL = 66
```

7 B0-1 Cycle-100 to Cycle-250 State-Initialized Continuation

B0-1 completed the Abaqus solver successfully. PBS reported exit status 2 only because the original postprocessing path failed after solver completion; extraction was rerun manually afterward.

Table 8: B0-1 run accounting and solver status.

Quantity	Value
PBS job	1336497.mmaster02
Route	cycle 100 state → cycle 250
PBS walltime	01:41:36
PBS CPU time	10:52:42
Requested CPUs	16
Abaqus solver	completed
Abaqus parallelism	1 MPI rank × 16 threads
Scientific status	Practical baseline accepted

Table 9: B0-1 comparison against the 1000-cycle reference at cycle 250.

Quantity	Relative error
`RF1_max`	0.259048%
`RF1_min`	0.687538%
`loop_area_abs`	0.967300%
`HOLE_RING_MISES_MAX`	7.63648%
`HOLE_RING_S11_MAX_ABS`	9.58753%
`HOLE_RING_SDV1_MAX`	0.360172%
`HOLE_RING_SDV8_MAX`	10.9164%
`HOLE_RING_SDV11_MAX`	7.27213%

The global response is excellent, but the local hole-ring stress and state-variable errors are too large for a clean exact-restart claim.

8 B0-1 Local Diagnostic Review

A pointwise hole-ring diagnostic was run to determine whether the high local error was only a maximum-location artifact. It was not. The reference and reinjection maxima for `SDV8` occurred at the same element and integration point.

Table 10: B0-1 SDV8 argmax diagnostic at cycle 250.

Quantity	Value
Reference SDV8 argmax	element 1242, IP 7
Reinjection SDV8 argmax	element 1242, IP 7
Same argmax location	true
Reference value	91.1466598511
Reinjection value	81.1967315674
Argmax relative error	10.9164%

This made B0-1 a real local mismatch rather than a harmless maximum-location shift.

9 B0 Cycle-100 Initialization-Only Audit

The initialization audit was created to answer whether the local mismatch already exists immediately after exact cycle-100 initialization, before any cycle-100 to cycle-250 continuation.

Table 11: Cycle-100 binary reader and initialization audit.

Quantity	Value
CSV-vs-binary maximum absolute error	3.55×10^{-15}
Common hole-ring element/IP records	480
`SDV8` median relative error	12.7022%
`SDV8` 95th percentile relative error	150.665%
`SDV8` maximum relative error	479.090%
`SDV8` argmax location	same element/IP, 1242/7

The binary extraction and reader pipeline is correct to machine precision. Therefore the local mismatch is not caused by the Python binary writer or direct-access file format. The mismatch appears during manual Abaqus initialization and the tiny equilibrium step.

9.1 Initialization Audit Interpretation

The most likely interpretation is that `SDVINI/SIGINI` inject stress and state variables correctly, but the manual state-initialized model does not reconstruct the compatible displacement, strain, solver-history, and full equilibrium state of the original reference analysis. Strain-like local state variables such as `SDV8` are therefore adjusted during the first equilibrium solve. This means B0-1 is useful as a practical state-initialized continuation test, but it is not a mathematically exact Abaqus restart for local fields.

10 Reinjection Error Floor and Revised Metric Policy

The B0 audit closes Stage 16N-B with the following conclusion:

> Global continuation works, binary state transfer works, but local exact restart is not achieved with `SDVINI/SIGINI` alone.

Future fixed and adaptive jump results must separate the manual reinjection error floor from the additional error caused by cycle-jump extrapolation. The primary pass/fail metrics are `RF1` maximum/minimum, loop area, global hysteresis shape, `HOLE_RING_SDV1_MAX`, `HOLE_RING_SDV11_MAX`, and `HOLE_RING_MISES_MAX`. `HOLE_RING_S11_MAX_ABS` remains important, but it is now treated as a diagnostic warning metric rather than a sole hard rejection variable, because the B1D3/B1D4/B1D5 boundary study showed non-monotonic local stress sensitivity. `HOLE_RING_SDV8_MAX` also remains diagnostic-only because it is already affected by initialization before any cycle-jump approximation is applied.

11 Stage 16N-C Fixed State-Initialized Jump Scaffold

Stage 16N-C now begins with one conservative fixed jump:

Table 12: First fixed state-initialized jump case.

Quantity	Value
Case name	`B1_100_to_125_to_250`
Base state	cycle 100
Interpreted jump target	cycle 125
Normal Abaqus continuation	cycles 126–250
Comparison endpoint	cycle 250
Initial state strategy	zero-order hold from cycle 100
Baseline for interpretation	B0-1 cycle 100 to 250 continuation

The zero-order state strategy is intentionally conservative. It measures the error added by skipping cycles 101–125 on top of the known manual reinjection floor. If this case gives acceptable global error and controlled primary local errors, the next fixed jump cases can be expanded to cycle 250 to 300, cycle 500 to 575, and cycle 750 to 850.

11.1 Prepared-Only B2 and B3 Cases

B2 and B3 may be prepared while B1 is running, but their solver jobs are gated by the B1 result. They must not be submitted until B1 has been extracted, compared against the 1000-cycle reference, and interpreted against the B0 reinjection baseline.

Table 13: Prepared-only fixed-jump cases held behind the B1 gate.

Case	Base state	Jump target	Compare endpoint
`B2_250_to_300_to_500`	cycle 250	cycle 300	cycle 500
`B3_500_to_575_to_750`	cycle 500	cycle 575	cycle 750

If B1 passes, B2 and B3 are the first paired submission under the new two-job policy. If B1 fails, the paired slots should instead be used for B1-small and B1-diagnostic variants.

11.2 B1 Result

B1 completed successfully as an Abaqus job, but it did not pass the fixed-jump method gate. The comparison at cycle 250 gave good global errors but unacceptable primary local state errors.

Table 14: B1 fixed-jump comparison against the 1000-cycle reference and B0 baseline.

Quantity	B1 total error	B0 baseline	Additional error
`RF1_max`	0.361433%	0.259048%	0.102384%
`RF1_min`	2.27018%	0.687538%	1.58264%
`loop_area_abs`	1.26223%	0.967300%	0.294934%
`HOLE_RING_MISES_MAX`	2.40141%	7.63648%	-5.23507%
`HOLE_RING_S11_MAX_ABS`	1.21869%	9.58753%	-8.36884%
`HOLE_RING_SDV1_MAX`	10.2916%	0.360172%	9.93139%
`HOLE_RING_SDV8_MAX`	16.1769%	10.9164%	5.26047%
`HOLE_RING_SDV11_MAX`	11.2777%	7.27213%	4.00560%

The B1 status is `review`, controlled by `HOLE_RING_SDV11_MAX`. Therefore B2 and B3 remain held. The two-job policy should next be used for smaller or diagnostic B1 variants rather than larger fixed jumps.

11.3 B1 Jump-Size Sensitivity

The next two-job batches reduced the first fixed jump from cycle 100 to smaller target cycles while keeping the same comparison endpoint at cycle 250. This directly tests whether the local hole-ring error decreases when fewer cycles are skipped.

Table 15: B1 fixed-jump sensitivity at cycle 250.

Case	Jump target	Skipped cycles	Status	Controlling quantity
`B1Q_100_to_106_to_250`	106	6	review, 5.75372%	`HOLE_RING_MISES_MAX`
`B1S_100_to_112_to_250`	112	12	review, 9.34203%	`HOLE_RING_S11_MAX_ABS`
`B1_100_to_125_to_250`	125	25	review, 11.2777%	`HOLE_RING_SDV11_MAX`

The sensitivity study confirms that the original jump from cycle 100 to 125 was too aggressive for local hole-ring variables. Reducing the target to cycle 106 reduced the maximum primary error to approximately 5.75%, but this remained above the strict 5% gate. The useful scientific conclusion is that the adaptive jump-size estimate near cycle 100 was too optimistic and must be reduced by an early-cycle safety factor. Since the reinjection-aware table suggested approximately $\Delta N \approx 24$ at cycle 100, the practical early safety factor is approximately

$$\frac{5}{24} \approx 0.21.$$

This supports an early-cycle safety-factor range of approximately 0.20–0.25, controlled by local field quantities rather than by `RF1` alone.

11.4 B1D3/B1D4/B1D5 Boundary Check

A stricter boundary check was then performed with target cycles 103, 104, and 105. The result is useful, but non-monotonic:

Table 16: B1D3/B1D4/B1D5 boundary check at cycle 250.

Case	Jump target	Skipped cycles	Status	Controlling quantity
`B1D3_100_to_103_to_250`	103	3	review, 9.13095%	`HOLE_RING_S11_MAX_ABS`
`B1D4_100_to_104_to_250`	104	4	review, 9.34307%	`HOLE_RING_S11_MAX_ABS`
`B1D5_100_to_105_to_250`	105	5	pass, 2.34986%	`HOLE_RING_SDV1_MAX`
`B1Q_100_to_106_to_250`	106	6	review, 5.75372%	`HOLE_RING_MISES_MAX`

B1D5 remains a useful successful case:

`100 -> 105 -> 250`, with maximum primary error 2.34986%.

However, B1D3 and B1D4 both failed under the local stress criterion even though they used shorter jumps. B1D3 was controlled by `HOLE_RING_S11_MAX_ABS` at 9.13095%, and B1D4 was controlled by the same metric at 9.34307%. Therefore the cycle-100 boundary is not robust under a hard local-stress maximum criterion. The isolated B1D5 pass should not be generalized as a universal accepted early-cycle rule.

11.5 B1D4 Pointwise Diagnostic

B1D4 was diagnosed at the hole-ring element/integration-point level using the 1000-cycle reference ODB and the B1D4 ODB at cycle 250. The diagnostic compared 480 common hole-ring element/IP records and checked both pointwise errors and argmax locations.

Table 17: B1D4 pointwise diagnostic summary at cycle 250.

Variable	Mean error	Median error	95th percentile	Max pointwise error
S11_ABS	13.8050%	11.6186%	35.2953%	121.4232%
MISES	8.05921%	–	–	–
SDV1	1.64737%	–	–	–
SDV8	131.337%	–	–	–
SDV11	50.4912%	–	–	–

For the controlling stress metric, the reference and B1D4 argmax location were the same:

> S11_ABS argmax element/IP = 1242/7 in both the reference and B1D4.

The global phase check did not show a cycle-phase mismatch. Both runs used CYCLE_0250; the cycle had local time from 0 to 1, displacement extrema matched exactly, and the global loop-area error was only 0.107074%. Therefore the B1D4 failure is not explained by a simple output-phase or step-numbering mismatch. It appears to be a real local stress/reinjection sensitivity in the B1D4 path.

This keeps the conservative decision unchanged: B1D5 is a useful pass case, but B1D3 and B1D4 show that the early-cycle jump boundary is not robust when local stress extrema are treated as hard primary rejection variables.

11.6 B1D3 Confirmation Result

B1D3 completed successfully as an Abaqus job and was compared against the 1000-cycle reference at cycle 250. It did not pass the fixed-jump gate.

Table 18: B1D3 comparison against the 1000-cycle reference at cycle 250.

Quantity	Relative error
RF1_max	1.45771%
RF1_min	0.661665%
loop_area_abs	0.214351%
HOLE_RING_MISES_MAX	7.43637%
HOLE_RING_S11_MAX_ABS	9.13095%
HOLE_RING_SDV1_MAX	1.52456%
HOLE_RING_SDV8_MAX	9.68802%
HOLE_RING_SDV11_MAX	5.43394%

B1D3 confirms that the B1D4 behavior is not an isolated single-case artifact. The global response remains accurate, but the local hole-ring stress maximum and SDV11 response exceed the strict local threshold. The current interpretation is therefore:

> Early-cycle jumping from cycle 100 is not robust under local stress-maximum acceptance. The pass at cycle 105 is useful evidence that small jumps can work, but the non-monotonic failures at cycles 103 and 104 require a more careful local acceptance strategy or a later-start jump strategy.

11.7 Revised Local-Stress Acceptance Policy

The B1D3/B1D4/B1D5 boundary is scientifically important because it shows that smaller jumps are not always better when judged by a single local stress maximum. The robust conclusion is:

> Global hysteresis quantities are robust, but local stress-maximum metrics near the hole are non-monotonic and highly sensitive to manual state initialization and re-equilibration.

Therefore, HOLE_RING_S11_MAX_ABS is retained and reported, but it should not be used as the only hard pass/fail metric. The revised three-level metric strategy is:

- Primary pass/fail: RF1 extrema, loop area, HOLE_RING_SDV1_MAX, HOLE_RING_SDV11_MAX, and HOLE_RING_MISES_MAX.
- Secondary/diagnostic: HOLE_RING_S11_MAX_ABS, single-point stress maxima, and SDV8.
- Robust local-field additions: mean/median, 95th percentile, or 99th percentile hole-ring stress errors from pointwise distributions.

This does not hide the local stress sensitivity. It reports it as evidence that local extrema are more restrictive and less stable than global hysteresis metrics.

The current reporting conclusion is:

> The B1 sensitivity study showed that the original jump from cycle 100 to 125 was too aggressive for the local hole-ring variables. Reducing the jump to cycle 105 gave an acceptable maximum primary error of 2.35%, but the cycle 103 and 104 cases both produced non-monotonic local stress errors controlled by HOLE_RING_S11_MAX_ABS. Therefore, early-cycle jumping from cycle 100 cannot yet be accepted as robust under a hard local stress-maximum criterion.

B2 and B3 remain held. The next two-job policy is therefore used for one smaller cycle-100 diagnostic jump and one later-cycle safe pilot, rather than for the original larger fixed jumps.

Table 19: Next two-job diagnostic pair submitted after the B1D3 result.

Case	Route	Purpose	PBS job
B1D2_100_to_102_to_250	100 → 102 → 250	cycle-100 hard-stress check	1341026.mmaster02
B2SAFE_250_to_260_to_500	250 → 260 → 500	later-start safe pilot	1341027.mmaster02
B2SAFE_EQ_250_to_260_to_500	250 → 260 → 500	smaller equilibration increment	1341028.mmaster02

The first B2SAFE submission failed after datacheck during FIXED_JUMP_EQUILIBRATE_TO_CYCLE_0260. Abaqus reported slow force convergence and TOO MANY ATTEMPTS MADE FOR THIS INCREMENT. A distinct rerun case, B2SAFE_EQ_250_to_260_to_500, reduced the equilibration initial increment from 1.0×10^{-6} to 1.0×10^{-8}, but it failed in the same first equilibration step after cutbacks to 6.25×10^{-10}. This is a useful result by itself: the later cycle-250 reinjection point is numerically more difficult to equilibrate than the cycle-100 cases, and it requires an explicit cycle-250 reinjection/equilibration diagnostic before B2-safe can be interpreted scientifically.

11.8 Unattended Dependency Queue

The unattended queue was reorganized into two dependency-controlled streams so that at most two 16-CPU jobs can run at once while preserving the scientific gates.

Table 20: Dependency-controlled unattended queue.

Stream	Case	PBS job	Dependency/status
A	B1D2_100_to_102_to_250	1341026.mmaster02	running
A	B1D1_100_to_101_to_250	1341029.mmaster02	afterany:1341026
B	B0_250_to_500	1341033.mmaster02	failed in reinjection equilibration
B	B2D2_250_to_252_to_500	1341034.mmaster02	afterok not satisfied
B	B2D5_250_to_255_to_500	1341035.mmaster02	afterok not satisfied
B	B0_AUDIT_0250_INITIALIZATION_ONLY	1341036.mmaster02	failed in equilibration

The first B0-2 exact-reinjection submission, 1341030.mmaster02, exited with status 126 because the PBS script had CRLF line endings and invoked `/bin/bash^M`. Its dependent jobs 1341031 and 1341032 did not run as meaningful solver jobs. The exact-reinjection submit writer was corrected to write LF-only PBS scripts, the remote PBS file was normalized, and Stream B was resubmitted as jobs 1341033–1341035.

The corrected B0-2 job, 1341033.mmaster02, then ran Abaqus and failed during the first `REINJECTION_EQUILIBRATE` increment with `TOO MANY ATTEMPTS MADE FOR THIS INCREMENT`. Therefore B2D2 and B2D5 correctly remained blocked by their `afterok` dependencies. The Stream B replacement, `B0_AUDIT_0250_INITIALIZATION_ONLY` job 1341036.mmaster02, also failed in its first initialization/equilibration increment after cutbacks to 6.25×10^{-11}. This means the cycle-250 problem is not a jump-size result: exact cycle-250 manual state injection cannot yet be equilibrated robustly.

11.9 Long-Weekend Dependency Extension

Because the user will be away for more than 40 hours, the unattended queue was extended with additional scientifically safe diagnostics. The queue still preserves the two-job policy: only one Stream A job and one Stream B job can run at a time, and every job requests 16 CPUs, 90 GB, and 24 hours.

Table 21: Extended long-weekend queue.

Stream	Case	PBS job	Dependency/status
A	B1D2_100_to_102_to_250	1341026.mmaster02	running
A	B1D1_100_to_101_to_250	1341029.mmaster02	afterany:1341026
A	B1D2_EQ_100_to_102_to_250	1341037.mmaster02	afterany:1341029
A	B1D1_EQ_100_to_101_to_250	1341038.mmaster02	afterany:1341037
B	B0_AUDIT_0500_INITIALIZATION_ONLY	1341039.mmaster02	running
B	B0_AUDIT_0250_SDVINI_ONLY	1341040.mmaster02	afterany:1341039
B	B0_AUDIT_0250_SIGINI_ONLY	1341041.mmaster02	afterany:1341040
B	B0_AUDIT_0500_SDVINI_ONLY	1341042.mmaster02	afterany:1341041
B	B0_AUDIT_0500_SIGINI_ONLY	1341043.mmaster02	afterany:1341042

The two `B1D*_EQ` jobs are fallback diagnostics with a smaller equilibration initial increment. They should be interpreted as numerical sensitivity checks for the cycle-100 boundary, not as authorization to submit B2/B3. The Stream B audit jobs split the manual cycle-250 and cycle-500 state injection into full, `SDVINI`-only, and `SIGINI`-only variants. This should identify whether the cycle-250/500 equilibration barrier is driven primarily by STATEV injection, stress injection, or the coupled manual restart.

The Stream B injection-only diagnostics completed quickly and all returned exit status 1. Therefore, a second B1-only EQ stream was submitted into the free slot. This keeps the cluster at the intended two active 16-core jobs while continuing to avoid B2/B3.

Table 22: Final long-weekend live and queued jobs.

Stream	Case	PBS job	Dependency/status
A	`B1D2_100_to_102_to_250`	1341026.mmaster02	running
A	`B1D1_100_to_101_to_250`	1341029.mmaster02	`afterany:1341026`
A	`B1D2_EQ_100_to_102_to_250`	1341037.mmaster02	`afterany:1341029`
A	`B1D1_EQ_100_to_101_to_250`	1341038.mmaster02	`afterany:1341037`
A	`B1D3_EQ_100_to_103_to_250`	1341044.mmaster02	`afterany:1341038`
A	`B1D4_EQ_100_to_104_to_250`	1341045.mmaster02	`afterany:1341044`
A	`B1D5_EQ_100_to_105_to_250`	1341046.mmaster02	`afterany:1341045`
C	`B1Q_EQ_100_to_106_to_250`	1341047.mmaster02	running
C	`B1S_EQ_100_to_112_to_250`	1341048.mmaster02	`afterany:1341047`
C	`B1_EQ_100_to_125_to_250`	1341049.mmaster02	`afterany:1341048`

The final live/queued solver count after this extension was 10 jobs: 2 running and 8 held by dependencies. The active requested resource load was therefore $2 \times 16 = 32$ CPU cores.

11.10 Post-Queue Check on 2026-06-07

The long-weekend queue had drained by the 2026-06-07 check: `qstat -u pr21vyci` returned no active jobs. The PBS extended-history query with `qstat -x -f` no longer exposed the listed job IDs, returning `Unknown Job Id` for 1341026, 1341029, 1341037, 1341038, 1341044–1341049, and the audit jobs 1341039–1341043. Therefore the final job assessment below is based on the completed run artifacts in the remote Stage 16N directory and the refreshed comparison CSVs, not on retained PBS accounting records.

Table 23: Final fixed-jump comparison after the dependency queue drained.

Case	Route	Status	Max primary error	Controller
`B1D1`	$100 \to 101 \to 250$	review	8.70868%	`SDV11`
`B1D1_EQ`	$100 \to 101 \to 250$	review	8.70868%	`SDV11`
`B1D2`	$100 \to 102 \to 250$	review	8.79420%	`SDV11`
`B1D2_EQ`	$100 \to 102 \to 250$	review	8.79420%	`SDV11`
`B1D3`	$100 \to 103 \to 250$	review	7.43637%	`MISES`
`B1D3_EQ`	$100 \to 103 \to 250$	review	7.43637%	`MISES`
`B1D4`	$100 \to 104 \to 250$	review	7.54718%	`SDV11`
`B1D4_EQ`	$100 \to 104 \to 250$	review	7.54718%	`SDV11`
`B1D5`	$100 \to 105 \to 250$	pass	2.34986%	`SDV1`
`B1D5_EQ`	$100 \to 105 \to 250$	pass	2.34986%	`SDV1`
`B1Q`	$100 \to 106 \to 250$	review	5.75372%	`MISES`
`B1Q_EQ`	$100 \to 106 \to 250$	review	5.75372%	`MISES`
`B1S`	$100 \to 112 \to 250$	review	7.63033%	`SDV11`
`B1S_EQ`	$100 \to 112 \to 250$	review	7.63033%	`SDV11`
`B1`	$100 \to 125 \to 250$	review	11.27770%	`SDV11`
`B1_EQ`	$100 \to 125 \to 250$	review	11.27770%	`SDV11`

The smaller-equilibration `EQ` reruns reproduced the same comparison errors as their non-`EQ` counterparts. This is important: the cycle-100 B1 jump-size trend is controlled by the physical/local state mismatch relative to the 1000-cycle reference, not by the initial equilibration increment used in the fixed-jump continuation. Under the current 5% primary-error gate, only the 100 → 105 → 250 case passes; the 100 → 106 and larger jumps remain above threshold, and the 100 → 101/102/103/104 cases still show local-state or Mises review errors.

The later-cycle fixed-jump candidates `B2`, `B2SAFE`, `B2SAFE_EQ`, `B2D2`, `B2D5`, and `B3` remain `missing_outputs`. This is expected because the cycle-250 exact-reinjection/equilibration gate failed and the unsafe dependent jobs were not allowed to produce accepted fixed-jump validations.

11.11 Robust Scalar-Metric Re-score

The completed B1/B1_EQ family was re-scored with robust scalar statistics: mean, median, 95th percentile, 99th percentile, and maximum relative error. This re-score uses the lightweight scalar metrics retained in the repository. It is not yet a full pointwise field-percentile study because complete per-integration-point fields were not retained for every B1 case. Loop errors were normalized by the full reference loop amplitude for `U1_avg` and `RF1_sum`, avoiding artificial blow-ups at zero crossings.

Table 24: Robust scalar-metric re-score for representative B1 cases.

Case	Mean	Median	p95	p99	Max	Status
`B1D1`	2.97163%	1.77699%	7.47991%	8.46293%	8.70868%	review
`B1D3`	2.78810%	1.49114%	6.93576%	7.33625%	7.43637%	review
`B1D5`	0.392238%	0.000247%	1.76313%	2.23251%	2.34986%	pass by max
`B1D5_EQ`	0.392238%	0.000247%	1.76313%	2.23251%	2.34986%	pass by max
`B1Q`	2.54352%	2.34673%	4.99948%	5.60287%	5.75372%	robust pass, max review
`B1Q_EQ`	2.54352%	2.34673%	4.99948%	5.60287%	5.75372%	robust pass, max review
`B1S`	3.65193%	3.12240%	7.50946%	7.60616%	7.63033%	review
`B1`	4.64409%	2.33580%	11.0312%	11.2284%	11.2777%	review

The robust scalar re-score supports the same conservative conclusion as the hard maximum-error gate. `B1D5` and `B1D5_EQ` are the only clean passes. `B1Q` and `B1Q_EQ` are scientifically interesting because their primary p95 error is just below 5%, but their maxima still exceed the 5% gate. Therefore they should be treated as borderline review cases unless a later full-field percentile acceptance rule is deliberately adopted.

11.12 Limitations of Manual SDVINI/SIGINI Reinjection

The later-cycle diagnostics show that the bottleneck is no longer only jump size. Manual `SDVINI/SIGINI` reinjection starts a scratch Abaqus model with injected stress and state variables, but it does not preserve the compatible displacement field, strain field, equilibrium state, solver history, or tangent/history information of the original inhomogeneous analysis. At cycle 100 this route is good enough for a narrow global-response demonstration. At cycle 250 and cycle 500 it failed before valid physical continuation in exact, `SDVINI`-only, `SIGINI`-only, and combined diagnostics.

This is a thesis-useful limitation result: manual state initialization is not robust enough for later-cycle reinjection in the plate-with-hole benchmark. Repeating B2/B3 jobs with the same scratch `SDVINI/SIGINI` route is therefore not expected to add scientific value. The next path is Stage 16N-R, a reinjection-strategy repair that preserves Abaqus' finite-element state through native restart or in-analysis continuation and applies the cycle jump only to independent material memory variables.

12 Failures and Fixes

Table 25: Important failures and fixes recorded in Stage 16N.

Issue	Observation	Fix / conclusion
Accidental serial-like run	A 22-hour run reached only 592 full cycles.	Parallel launcher corrected; later 1000-cycle reference completed.
30-CPU over-allocation	30-thread run averaged about 10 effective cores.	16 CPUs selected as production compromise.
Missing start/end email	PBS accounting showed `Mail_Points = a`.	Added `#PBS -m abe` and mail user to Stage 16N submit scripts.
CSV state-reader race	Threaded Abaqus calls attempted concurrent CSV preload.	Replaced with direct-access binary state reader.
Wrong direct-access record length	Abaqus/Intel runtime expected word-based `RECL`.	Set `RECL = 66` for 33 double values.
Postprocessing path failure	B0-1 solver completed but PBS exit status was 2.	Fixed extractor path and reran extraction manually.
B0-1 local mismatch	Global errors under 1%, local `SDV8` max error 10.9164%.	Added pointwise diagnostic and initialization-only audit.
Initialization local mismatch	Binary audit passed, but `SDV8` mismatch appeared immediately.	Concluded manual `SDVINI/SIGINI` is not a mathematically exact local restart.
Metric policy mismatch	Original adaptive table allowed `SDV8` to act as a hard controller.	Added reinjection-aware table with `SDV8` diagnostic-only.
B1 fixed-jump gate	Global B1 errors were small, but `SDV1` and `SDV11` primary local errors exceeded 10%.	Hold B2/B3 and use the two-job policy for smaller/debug B1 variants.
B1D4/B1D5 non-monotonic boundary	B1D5 passed at 2.34986%, but B1D4 failed at 9.34307% controlled by `S11`.	Accept `100 -> 105` only with diagnostic confirmation; diagnose B1D4 before B2/B3.
B1D4 pointwise diagnostic	B1D4 and the reference share the same `S11_ABS` argmax location, and median/p95 stress errors are not negligible.	Treat B1D4 as real local stress sensitivity, not a simple phase or argmax-swap artifact.
B1D3 confirmation	B1D3 also failed at 9.13095%, controlled by `S11_ABS`.	Do not generalize the B1D5 pass; treat cycle-100 jumping as not robust under hard local stress maxima.
Metric policy revision	`S11_MAX_ABS` is non-monotonic across B1D3/B1D4/B1D5.	Keep `S11` maxima as diagnostic warnings; use global, STATEV, and Mises-type metrics as primary acceptance controls.

Issue	Observation	Fix / conclusion
Completed B1 dependency queue	The drained queue shows `B1D5` and `B1D5_EQ` pass at 2.34986%, while all other completed B1-size variants remain review cases.	Treat 100 $\rightarrow$ 105 $\rightarrow$ 250 as the only currently accepted B1 fixed jump; keep B2/B3 blocked by the cycle-250 reinjection gate.
Cycle-250 reinjection gate	B2SAFE, B2SAFE_EQ, exact B0-2, and `B0_AUDIT_0250_INITIALIZATION_ONLY` all failed in their first equilibration increment.	Hold all cycle-250 fixed jumps; diagnose the cycle-250 manually injected stress/state field before rerunning B2SAFE.

13 Current Conclusion

Stage 16N has successfully produced a full 1000-cycle non-jump reference and a conservative adaptive ΔN table. The inhomogeneous plate-with-hole benchmark is scientifically useful because both global hysteresis quantities and local hole-ring state variables evolve significantly. The HPC workflow is resource-controlled with a 16-CPU production policy and explicit start/end/failure mail notifications.

The fixed-jump result is now frozen as Stage 16N-C. The cycle-jump method can work in the inhomogeneous Abaqus model, but only for a very conservative early jump: `B1D5`/`B1D5_EQ`, 100 $\rightarrow$ 105 $\rightarrow$ 250, with 2.34986% maximum primary scalar-metric error. The adaptive cycle-100 table estimated $\Delta N \approx 24$, while the accepted observed jump is $\Delta N = 5$, giving an effective safety factor of about $5/24 \approx 0.21$. Thus the thesis-safe result is that adaptive ΔN estimates must be strongly reduced when applied to the inhomogeneous plate-with-hole benchmark. Local hole-ring stress and STATEV extrema are much more restrictive than the global hysteresis response.

The important methodological limitation is exact local reinjection. B0-1 proves that the manual `SDVINI`/`SIGINI` workflow can run and reproduce the global force-displacement response well near cycle 100, but it does not reproduce local strain-like state variables exactly. The cycle-100 initialization-only audit shows that this local mismatch appears immediately during manual Abaqus initialization/equilibration, even though the binary reader is exact. The cycle-250 and cycle-500 exact, `SDVINI`-only, `SIGINI`-only, and combined diagnostic failures show that the same mechanism becomes a hard equilibration gate at later cycles. The next scientific path is therefore not more scratch state-initialized B2/B3 jobs, but Stage 16N-R: a restart-preserved material-state jumping strategy.

14 Next Work

The completed dependency queue shows that a small jump from cycle 100 can work only in a narrow window: `B1D5_100_to_105_to_250` and `B1D5_EQ_100_to_105_to_250` passed with maximum primary error 2.34986%. The surrounding smaller and larger B1 variants remain review cases, including `B1D1`, `B1D2`, `B1D3`, `B1D4`, `B1Q`, `B1S`, and `B1`. The `EQ` fallback cases did not reduce the comparison errors. B0-2 and B0-3 remain on hold as exact-local-restart claims, and B2/B3 fixed jumps are also held because the cycle-250 reinjection/equilibration gate has not passed. The next work is the Stage 16N-R repair path.

1. Freeze Stage 16N-C as a validated small-jump demonstration plus a limitation study of manual reinjection.

2. Do not submit B2/B3/B3SAFE/B4SAFE with the current scratch `SDVINI/SIGINI` route.
3. Use the robust scalar re-score as supporting evidence: only `B1D5` and `B1D5_EQ` pass the hard 5% maximum primary-error gate; `B1Q/B1Q_EQ` remain borderline review cases.
4. Treat the completed STATEV independence audit as the first Stage 16N-R artifact. Independent memory variables are `STATEV(1)`, `STATEV(2:19)`, and `STATEV(20:25)`; `STATEV(26)` is derived from alpha and `STATEV(27)` is increment-local/diagnostic.
5. Prepare a restart-enabled reference or checkpoint run because no local usable `.res/.stt/.mdl/.sim` restart files were found for the completed Stage 16N reference.
6. Run a native Abaqus restart control before any new jump: continue from checkpoint without memory overwrite and compare 250 $\rightarrow$ 500 and 500 $\rightarrow$ 1000 against the full reference.
7. Implement restart-preserved material-memory overwrite only after the native restart control passes, and modify only the independent material memory variables inside the UMAT.
8. Then retry later jumps with the preserved FE state, starting with small controlled jumps rather than the original B2/B3 queue.

15 Stage 16N-R1 Restart-Enabled Reference Submission

Stage 16N-R1 was started on 2026-06-07 to create native Abaqus restart files before any restart-preserved UMAT overwrite is attempted. Two ordinary no-jump reference jobs were generated and submitted under the two-job policy:

Table 26: Stage 16N-R1 restart-enabled reference jobs.

Case	PBS job	Abaqus job	Target	Restart checkpoints
`R1B`	`1341177.mmaster02`	`stage16n_r1b_restart_ref_250cycles`	250 cycles	100, 250
`R1A`	`1341178.mmaster02`	`stage16n_r1a_restart_ref_500cycles`	500 cycles	100, 250, 500

Both jobs request `select=1:ncpus=16:mpiprocs=1:ompthreads=16:mem=90gb`, `walltime=24:00:00`, and run Abaqus with `cpus=16 mp_mode=threads`. The first attempted submissions, `1341175.mmaster02` and `1341176.mmaster02`, failed before Abaqus started because the generated per-case shell runner had Windows CRLF line endings. The runner was corrected to write LF line endings explicitly, the remote copies were normalized, and the corrected jobs were submitted as `1341177.mmaster02` and `1341178.mmaster02`. At the first `qstat -f` check both corrected jobs were running.

The acceptance condition for this substage is not a cycle-jump error threshold. It is the existence of usable native Abaqus restart files from the 250-cycle and 500-cycle checkpoint references. Only after those restart sources exist should native restart continuation tests be prepared.

15.1 Stage 16N-R1 Completion Check on 2026-06-08

Both restart-enabled reference analyses completed successfully as Abaqus jobs. PBS marked the jobs with exit status 2 because the post-run status-file shell block failed after extraction; this was a scripting issue, not a solver failure. The `.sta` files for both jobs end with `THE ANALYSIS HAS COMPLETED SUCCESSFULLY`, and both logs show `1 MPI RANK x 16 THREADS`.

Table 27: Stage 16N-R1 completed job accounting.

Case	PBS job	Walltime	CPU time	CPU percent	PBS status
R1B	1341177.mmaster02	03:13:22	18:58:17	655	exit 2 after extraction
R1A	1341178.mmaster02	08:32:36	43:30:54	661	exit 2 after extraction

Native restart files now exist. The 250-cycle reference produced `.res` (29 MB), `.mdl` (50 MB), `.sim` (1.6 MB), and `.stt` (226 GB). The 500-cycle reference produced `.res` (78 MB), `.mdl` (107 MB), `.sim` (1.6 MB), and `.stt` (1.1 TB). These large `.stt` files should not be deleted until native restart continuation has been tested.

The extracted endpoint global metrics match the existing 1000-cycle reference exactly at the checked cycles. For `R1B` at cycle 250 and `R1A` at cycle 500, the shared `U1_max`, `U1_min`, `RF1_max`, `RF1_min`, and `loop_area_abs` errors are all 0%. Therefore Stage 16N-R1 achieved its immediate purpose: FE-consistent native restart source files now exist and the restart-enabled no-jump runs reproduce the existing reference metrics at the endpoints.

The next substage should be native restart continuation control with no UMAT overwrite: restart from cycle 100 and continue to 250, then restart from cycle 250 and continue to 500. Only after those native restart continuations pass should restart-preserved UMAT memory overwrite be implemented.

16 Stage 16N-R2 Native Restart Control Submission

Stage 16N-R2 was prepared on 2026-06-08 as the next gate after the restart-enabled R1 references. The purpose is to test native Abaqus restart continuation with no `SDVINI/SIGINI` scratch reinjection and no UMAT memory overwrite. This directly checks whether Abaqus can preserve the finite-element displacement, strain, equilibrium, solver-history, and material state from its own restart files.

Table 28: Stage 16N-R2 native restart control jobs.

Case	PBS job	Restart read	Continuation	Scientific status
R2C1	1341284.mmaster02	step 100, inc 53	$100 \rightarrow 250$	pass after follow-up postprocess
R2C2	1341285.mmaster02	step 250, inc 58	$250 \rightarrow 500$	pass after follow-up postprocess

Both corrected jobs request `select=1:ncpus=16:mpiprocs=1:ompthreads=16:mem=90gb`, `walltime=24:00:00`, and run Abaqus with `cpus=16 mp_mode=threads`. The first submitted R2 attempt, `1341282.mmaster02` and `1341283.mmaster02`, failed during Abaqus datacheck because the oldjob `.odb` file was not symlinked into the restart case folders. This was a restart-setup issue, not a solver result. The link script was corrected to expose the oldjob `.odb` together with `.res`, `.stt`, `.mdl`, `.sim`, and `.prt`; the corrected submissions then ran to completion.

16.1 Stage 16N-R2 Completion Check on 2026-06-08

PBS history shows that both native restart control jobs finished with `Exit_status = 1`; however, both Abaqus solver runs completed successfully. Follow-up extraction and comparison then completed on HPC, and the scientific restart-control gate passed.

Table 29: Stage 16N-R2 finished job accounting.

Case	Walltime	CPU time	CPU percent	Mem	Vmem	Finish time
R2C1	01:55:33	11:15:42	639	94374872kb	5116372kb	Mon Jun 8 08:42:46 2026
R2C2	03:43:52	20:26:04	643	94375848kb	8489656kb	Mon Jun 8 10:31:06 2026

The full PBS accounting for `R2C1` records `ctime = Mon Jun 8 06:47:01 2026`, `stime = Mon Jun 8 06:47:09 2026`, `exec_host = mnode100/0*0`, and `Exit_status = 1`. The full PBS accounting for `R2C2` records `ctime = Mon Jun 8 06:47:01 2026`, `stime = Mon Jun 8 06:47:07 2026`, `exec_host = mnode101/0*0`, and `Exit_status = 1`. The nonzero PBS status is classified as wrapper/postprocessing failure, not Abaqus restart failure.

16.2 Stage 16N-R2 Follow-up Comparison

Manual follow-up postprocessing produced `stage16n_native_restart_comparison_summary.csv` in both R2 case folders. The `R2C1` comparison at cycle 250 reports `status = pass`, `max_global_error_pct = 0`, and `max_primary_local_error_pct = 0`. The `R2C2` comparison at cycle 500 reports the same values. This confirms that native Abaqus restart can preserve the inhomogeneous finite-element state and reproduce the 1000-cycle reference exactly at the checked continuation end-points.

17 Stage 16N-R3 Restart Debug Smoke Tests

Stage 16N-R3D was run as a minimal native restart diagnostic after the R2 wrapper/postprocessing failure. The aim was to preserve the Abaqus finite-element restart state and inspect the first UMAT calls after restart before implementing any material-memory overwrite.

Table 30: Stage 16N-R3D restart debug jobs.

Case	PBS job	Restart	Target	Solver	Debug	Exit
R3D1	1341601.mmaster02	250	251	completed	pass	0
R3D2	1341602.mmaster02	500	501	completed	pass	0

PBS history was checked on 2026-06-09. Both jobs used `select=1:ncpus=16:mpiprocs=1:ompthreads=16:`, `Resource_List.ncpus = 16`, `Resource_List.mpiprocs = 1`, `Resource_List.mem = 90gb`, and `Resource_List.walltime = 24:00:00`. `R3D1` ran on `mnode100/0*0`, with `ctime = Mon Jun 8 12:35:10 2026`, `stime = Mon Jun 8 12:35:12 2026`, `mtime = Mon Jun 8 12:36:22 2026`, `resources_used.walltime = 00:01:04`, `resources_used.cput = 00:04:30`, `resources_used.cpupercent = 422`, `resources_used.mem = 1545812kb`, `resources_used.vmem = 4859968kb`, `resources_used.ncpus = 16`, and `Exit_status = 0`. `R3D2` ran on `mnode101/0*0`, with `ctime = Mon Jun 8 12:35:11 2026`, `stime = Mon Jun 8 12:35:12 2026`, `mtime = Mon Jun 8 12:36:45 2026`, `resources_used.walltime = 00:01:26`, `resources_used.cput = 00:06:41`, `resources_used.cpupercent = 461`, `resources_used.mem = 2125400kb`, `resources_used.vmem = 8032600kb`, `resources_used.ncpus = 16`, and `Exit_status = 0`.

The restart debug traces show that the first safe overwrite hook is the restart continuation step with `KINC = 0`, `TIME(1) = 0`, and `TIME(2)` equal to the checkpoint cycle within floating-point tolerance. For `R3D1`, the first restart call had `JSTEP(1) = 251`, `KINC = 0`, and `TIME(2) = 250.000000000019`. For `R3D2`, the first restart call had `JSTEP(1) = 501`, `KINC = 0`, and `TIME(2) = 499.999999999974`. This confirms that native restart exposes the restarted material state to UMAT before the first continuation increment.

The next thesis-safe implementation step is Stage 16N-R3E exact no-op overwrite. The overwrite should target only independent material memory variables, namely `STATEV(1)`, `STATEV(2:19)`, and `STATEV(20:25)`. `STATEV(26)` and `STATEV(27)` should not be independently overwritten because they are derived or diagnostic state.

18 Stage 16N-R3E Exact Restart-Preserved Overwrite

Stage 16N-R3E is the first active implementation of the restart-preserved route. It is not yet a cycle-jump test. The objective is to prove that a UMAT can overwrite exact material memory inside a native Abaqus restart run without destroying the preserved displacement, strain, equilibrium, and solver state.

The prepared controls are:

Table 31: Stage 16N-R3E exact overwrite controls submitted on 2026-06-09.

Case	PBS job	Restart	Target	Overwrite trigger	Variables
`R3E1`	`1342248.mmaster02`	250	500	`JSTEP(1)=251`, `KINC=0`	`STATEV(1:25)`
`R3E2`	`1342249.mmaster02`	500	750	`JSTEP(1)=501`, `KINC=0`	`STATEV(1:25)`

The UMAT hook reads exact per-element/per-integration-point state from the restart-source `.odb`, converted to a direct-access binary table. It overwrites only independent material-memory variables, `STATEV(1:25)`. It does not table-overwrite `STATEV(26)` or `STATEV(27)`; `STATEV(26)` remains derived by the constitutive update and `STATEV(27)` remains increment-local diagnostic state.

The first R3E submission attempt, `1342246.mmaster02` and `1342247.mmaster02`, failed before the solve because Abaqus could not resolve `scratch=.` from the long case directory. This was an infrastructure issue, not a UMAT or restart-state result. The runner was corrected to use the short PBS scratch directory `$PBS_JOBDIR`. Corrected submissions `1342248.mmaster02` and `1342249.mmaster02` started at Tue Jun 9 06:09:58 2026. Both request `select=1:ncpus=16:mpiprocs=1:ompth` `walltime=24:00:00`, and both datachecks compiled the overwrite UMAT successfully before entering the production solve.

Both corrected jobs completed Abaqus successfully. PBS history reports `Exit_status=1` and `Stageout_status=1` for both jobs, but the solver files show `THE ANALYSIS HAS COMPLETED SUCCESSFULLY` in the `.sta` files and `THE ANALYSIS HAS BEEN COMPLETED` in the `.msg` files. The nonzero PBS status is therefore a stage-out or wrapper artifact, not a solver failure. Final accounting was:

Table 32: Stage 16N-R3E final PBS accounting.

Job	Case	Walltime	CPU time	CPU %	Memory	Host
`1342248`	R3E1	04:06:17	22:36:56	639	94375880kb	`mnode101`
`1342249`	R3E2	04:10:00	22:43:25	643	94375816kb	`mnode102`

The comparison against the reference was exact at the target cycles:

Table 33: Stage 16N-R3E exact overwrite comparison results.

Case	Target cycle	Status	Max global error	Max primary local error
R3E1	500	pass	0%	0%
R3E2	750	pass	0%	0%

R3E1 was compared against the default Stage 16N 1000-cycle pilot reference at cycle 500. R3E2 was compared against the parallel 1000-cycle reference because the default pilot comparison table did not contain cycle 750. The exact zero-error result confirms that modifying only independent material-memory variables inside a native restart does not destroy the preserved finite-element equilibrium state.

The current thesis-safe conclusion is therefore:

> Native Abaqus restart reproduced the reference exactly, proving that the inhomogeneous FE state is restartable when Abaqus preserves the displacement, strain, and equilibrium state. The next step is therefore to apply state changes inside the restarted analysis, rather than attempting scratch initialization through `SDVINI/SIGINI`.

19 Stage 16N-R3J Restart-Preserved Jump Tests

Stage 16N-R3J is the first nonzero repaired cycle-jump workflow after the R3E exact-overwrite pass. The finite-element displacement, strain, equilibrium, and solver state are still preserved by native Abaqus restart. The UMAT modifies only independent material memory, `STATEV(1:25)`, at the restart continuation call with `KINC=0`. The state modification is a cycle-space extrapolation,

$$\texttt{STATEV\_jump} = \texttt{STATEV\_base} + \Delta N \frac{d\texttt{STATEV}}{dN}.$$

The active production submissions are:

Table 34: Stage 16N-R3J active restart-preserved jump jobs submitted on 2026-06-10.

Case	PBS job	Restart	Slope pair	Material jump	Target
`R3J1`	`1342923.mmaster02`	250	100 → 250	250 → 255	500
`R3J2`	`1342924.mmaster02`	500	250 → 500	500 → 505	750

The originally desired one-cycle slope pairs, 249 → 250 and 499 → 500, were not available in the R1A restart-source `.odb` as stress/SDV field-output frames. The active corrected jobs therefore use the nearest available endpoint field-output pairs ending at the restart checkpoints: 100 → 250 for R3J1 and 250 → 500 for R3J2. This slope choice is less local than the initial ideal one-cycle slope, so it must be reported with the R3J results.

Several setup-only attempts were needed before the active production jobs. Jobs `1342915` and `1342916` failed because an upload quoting issue populated incorrect remote directories. Jobs `1342917` and `1342918` failed because cycles 249 and 499 had no stress/SDV field-output frames. Jobs `1342919` and `1342920` failed because the cluster Python rejected `from __future__ import annotations`. Jobs `1342921` and `1342922` failed because the older `pathlib.write_text` implementation rejected the `newline` keyword. These were setup and compatibility failures before the production solve, not scientific R3J failures.

The active corrected submissions, `1342923.mmaster02` and `1342924.mmaster02`, passed state extraction, extrapolated-state creation, UMAT compile/link, and Abaqus datacheck. At the first live resource check both were running under the standard production setup: `select=1:ncpus=16:mpipr` `walltime=24:00:00`, and Abaqus `cpus=16 mp_mode=threads`. `1342923` ran on `mnode101/0*0`; `1342924` ran on `mnode102/0*0`.

The R3J comparison classifies results using global RF/U/loop metrics, primary local metrics `HOLE_RING_MISES_MAX`, `HOLE_RING_SDV1_MAX`, `HOLE_RING_SDV8_MAX`, and `HOLE_RING_SDV11_MAX`, and diagnostic-only `HOLE_RING_S11_MAX_ABS`. The pass threshold is `max_primary_local_error_pct <= 5`; 5–10% is review; larger than 10% or solver instability is fail.

19.1 R3J Final Check on 2026-06-11

On 2026-06-11, PBS history showed that both active R3J jobs had finished. PBS reported `Exit_status=1` and `Stageout_status=1` for both jobs, but the Abaqus solver runs completed successfully. In both cases, the `.sta` file ends with `THE ANALYSIS HAS COMPLETED SUCCESSFULLY`. The nonzero PBS status came from the wrapper comparison step using Windows-style reference paths during the original job, not from solver failure.

Table 35: Stage 16N-R3J final PBS accounting.

Job	Case	Walltime	CPU time	CPU %	Memory	Host
1342923	R3J1	04:07:53	22:42:03	637	94375924kb	mnode101
1342924	R3J2	04:10:06	22:44:30	646	94377920kb	mnode102

Both jobs requested `select=1:ncpus=16:mpiprocs=1:ompthreads=16:mem=90gb` with `walltime=24:00:0` R3J1 started at Wed Jun 10 16:58:01 2026 and finished at Wed Jun 10 21:05:59 2026. R3J2 started at Wed Jun 10 16:58:00 2026 and finished at Wed Jun 10 21:08:11 2026.

The comparison was rerun manually on HPC with Linux reference paths. R3J1 was compared against the pilot 1000-cycle reference at cycle 500, and R3J2 was compared against the parallel 1000-cycle reference at cycle 750. Both restart-preserved jump tests passed exactly:

Table 36: Stage 16N-R3J restart-preserved jump comparison results.

Case	Target cycle	Status	Max global error	Max primary local error	Diagnostic S11 error
R3J1	500	pass	0%	0%	0%
R3J2	750	pass	0%	0%	0%

This is the first successful nonzero repaired cycle-jump gate for the inhomogeneous plate-with-hole benchmark. Within the checked metrics, native restart plus independent material-memory extrapolation preserved the FE state and reproduced the reference exactly after the +5 material-state jump.

19.2 R3J Zero-Error Audit

Because the R3J result is exactly zero for a nonzero material-state extrapolation, a targeted audit was performed on 2026-06-11 before moving to larger jumps. The audit confirms that the comparison did not accidentally compare a reference file against itself. The R3J1 jump metrics CSV and pilot reference metrics CSV had different inode numbers, sizes, and SHA-256 hashes. The R3J2 jump metrics CSV and parallel reference metrics CSV were likewise distinct.

The extrapolated state files also differ from the base restart states. For R3J1, all 25,184 integration-point records changed between cycle 250 and the extrapolated cycle-255 state; 478,496 state values changed, with maximum absolute delta 0.527609507243. For R3J2, all 25,184 records changed between cycle 500 and the extrapolated cycle-505 state; 478,496 state values changed, with maximum absolute delta 0.294372863770.

The UMAT overwrite trace was present in the Abaqus `.dat` files, not the `.msg` files. R3J1 printed `STAGE16N_R3J_OVERWRITE` at `KSTEP=251`, `KINC=0`, and `TIME(2)=250.000000000019`. R3J2 printed the same marker at `KSTEP=501`, `KINC=0`, and `TIME(2)=499.999999999974`. This confirms that the jump was applied once at the intended native-restart continuation hook. The generator was corrected so future wrapper logs grep the `.dat` file for this marker.

The endpoint cycle accounting was also correct. R3J1 produced 250 continuation rows, cycles 251–500, and compared against reference cycle 500. R3J2 produced 250 continuation rows, cycles 501–750, and compared against reference cycle 750. The comparison details for both cases report zero error in `U1`, `RF1`, loop area, `HOLE_RING_MISES_MAX`, `HOLE_RING_SDV1_MAX`, `HOLE_RING_SDV8_MAX`, `HOLE_RING_SDV11_MAX`, and diagnostic `HOLE_RING_S11_MAX_ABS`.

The thesis-safe interpretation is therefore:

> The restart-preserved cycle-jump strategy was successfully demonstrated for nonzero +5 cycle jumps. Both jumps, from cycle 250 to 255 and from cycle 500 to 505, completed successfully in Abaqus and reproduced the corresponding full reference endpoints with zero global and primary local error. This confirms that the previous reinjection problem was not a limitation of the material model or the inhomogeneous plate-with-hole geometry, but of the scratch `SDVINI/SIGINI` initialization strategy.

19.3 R3J +10 Cases Completed

The next conservative fixed-jump cases, R3J3 and R3J4, were submitted on 2026-06-11 and completed successfully. These jobs test whether the restart-preserved material-memory jump remains exact when the extrapolated material-state offset is doubled from +5 cycles to +10 cycles.

Table 37: Stage 16N-R3J +10 restart-preserved jump cases.

Case	PBS job	Restart	Slope pair	Material jump	Target
`R3J3`	`1344231.mmaster02`	250	100 → 250	250 → 260	500
`R3J4`	`1344232.mmaster02`	500	250 → 500	500 → 510	750

PBS history reported `Exit_status=0` for both jobs, while `Stageout_status=1` remained present. The Abaqus solver status is therefore successful, and the stage-out warning is treated as a file-transfer/accounting issue rather than a solver failure. The `.sta` files for both jobs end with `THE ANALYSIS HAS COMPLETED SUCCESSFULLY`.

Table 38: Stage 16N-R3J +10 final PBS accounting.

Job	Case	Walltime	CPU time	CPU %	Memory	Host
`1344231`	R3J3	04:08:42	22:51:08	646	94375928kb	`mnode101`
`1344232`	R3J4	04:07:59	22:41:15	656	94377792kb	`mnode102`

Both jobs requested `select=1:ncpus=16:mpiprocs=1:ompthreads=16:mem=90gb` with `walltime=24:00:0` R3J3 started at Thu Jun 11 09:55:29 2026 and finished at Thu Jun 11 14:04:16 2026. R3J4 started at Thu Jun 11 09:55:29 2026 and finished at Thu Jun 11 14:03:34 2026.

Table 39: Stage 16N-R3J +10 comparison results.

Case	Target cycle	Status	Max global error	Max primary local error	Diagnostic S11 error
R3J3	500	pass	0%	0%	0%
R3J4	750	pass	0%	0%	0%

The +10 results preserve the same zero-error endpoint behavior observed in the +5 cases. Within the current comparison metrics, the restart-preserved method has now passed nonzero fixed jumps of +5 and +10 cycles at both restart checkpoints. The recommended progression is +20 next, and adaptive ΔN restart-preserved jumping only after the fixed-jump accuracy boundary is known.

Lightweight R3J3/R3J4 evidence was copied into the repository. Heavy Abaqus result and restart files, including `.stt`, `.odb`, `.res`, `.sim`, `.mdl`, `.prt`, large `.msg`, and large `.dat` files, are intentionally excluded from GitHub and are being offloaded from `/home` to `/scratch`.

A Key Repository Evidence Files

- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_parallel_ma`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_adaptive_de`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_adaptive_de`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_REINJECTION`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_FIXED_JUMP_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_make_b1_rob`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_STATEV_INDE`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_audit_umat_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_NATIVE_REST`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/prepare_stage16n_res`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/run_stage16n_native_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_restart_con`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_restart_con`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_restart_con`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/prepare_stage16n_r3_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_compare_nat`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_restart_con`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_restart_con`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_restart_con`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_PROFESSOR_U`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
- `runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_`
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- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_fixed_jump_
- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj
- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj
- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj
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- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj
- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/stage16n_exact_reinj
- runs/chaboche_umat/stage16_abaqus_inhomogeneous_cycle_jump_benchmark/STAGE16N_HPC_MAIL_PO